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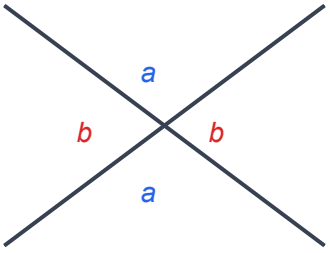
4.1 Properties of Shapes and Angles

OCR GCSE Mathematics J560 | Component 4 (Geometry and Measures) | Papers 1, 2 & 3

4.1.1 Basic Angle Rules

Rule	Statement	Diagram description
Angles on a straight line	Sum = 180°	Two or more angles forming a straight line
Angles around a point	Sum = 360°	All angles meeting at a single point
Vertically opposite angles	Equal	Two intersecting lines form two pairs of equal angles
Supplementary angles	Sum = 180°	Any two angles that add to 180°
Complementary angles	Sum = 90°	Any two angles that add to 90°



Vertically opposite: a = a, b =

Fig 1 — Vertically opposite angles are equal

Worked Example: Two angles on a straight line are $(3x + 20)^\circ$ and $(x + 40)^\circ$. Find x.

$$(3x + 20) + (x + 40) = 180$$
$$4x + 60 = 180$$

$4x = 120$
$x = \mathbf{30}$

4.1.2 Angles and Parallel Lines

When a **transversal** crosses two parallel lines, three types of angle pairs are formed. You must know their names, their letter shapes, and whether they are equal or supplementary.

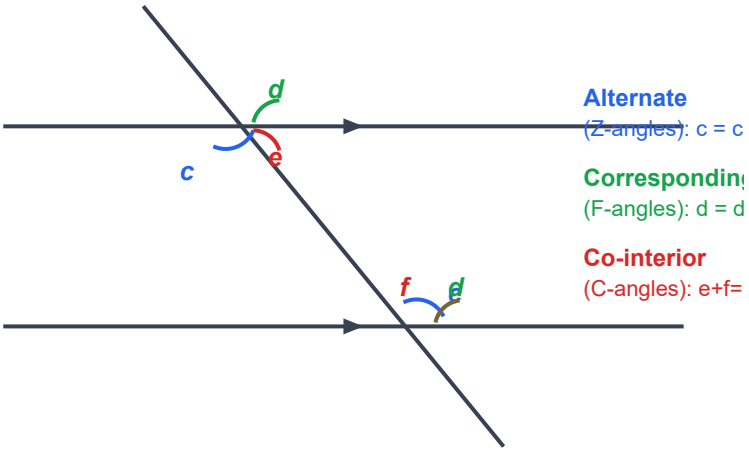


Fig 2 — Parallel lines with transversal: alternate (blue, equal), corresponding (green, equal), co-interior (red, supplementary)

Angle Pair	Letter Shape	Relationship	Reason to write
Alternate angles	Z (or N)	Equal	"alternate angles, parallel lines"
Corresponding angles	F	Equal	"corresponding angles, parallel lines"
Co-interior angles	C (or U)	Sum = 180°	"co-interior angles, parallel lines"

Exam Tip: Always state the reason when finding angles from parallel lines. "Z angles" alone is not sufficient — write "alternate angles, parallel lines".

Worked Example: Lines AB and CD are parallel. A transversal creates an angle of 65° with AB. Find the co-interior angle with CD.

Co-interior angles sum to 180°.
Co-interior angle = $180^\circ - 65^\circ = \mathbf{115^\circ}$
Reason: co-interior angles, AB parallel to CD.

4.1.3 Angles in Triangles and Quadrilaterals

Triangle: angles sum to 180°
Quadrilateral: angles sum to 360°

Types of Triangle

Triangle	Properties
Equilateral	All sides equal; all angles = 60°
Isosceles	Two sides equal; base angles equal
Scalene	All sides different; all angles different
Right-angled	One angle = 90°
Obtuse	One angle $> 90^\circ$
Acute	All angles $< 90^\circ$

The **exterior angle** of a triangle equals the sum of the two non-adjacent interior angles (the "remote interior angles").

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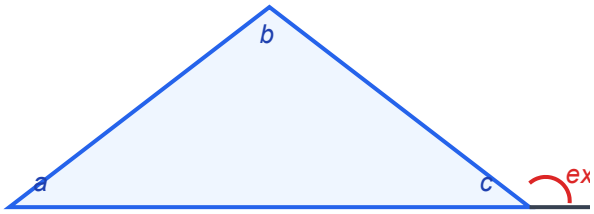


ext = a + b (exterior angle theorem)

Fig 3 — Exterior angle of a triangle = sum of the two remote interior angles

Worked Example: In triangle PQR, angle P = 48° and angle Q = 67°. Find the exterior angle at R.

Interior angle R = $180 - 48 - 67 = 65^\circ$
Exterior angle at R = $180 - 65 = 115^\circ$
Or directly: exterior angle = $48 + 67 = \mathbf{115}^\circ$

4.1.4 Angles in Polygons

Sum of interior angles of a polygon with n sides:

$$S = (n - 2) \times 180^\circ$$

Sum of exterior angles of any convex polygon = 360° (always)

For a **regular polygon** with n sides:

$$\text{Each interior angle} = \frac{(n-2) \times 180^\circ}{n}$$

$$\text{Each exterior angle} = \frac{360^\circ}{n}$$

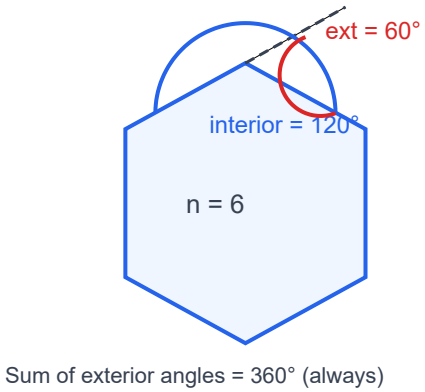


Fig 4 — Regular hexagon: interior angle = 120°, exterior angle = 60°, sum of all exterior angles = 360°

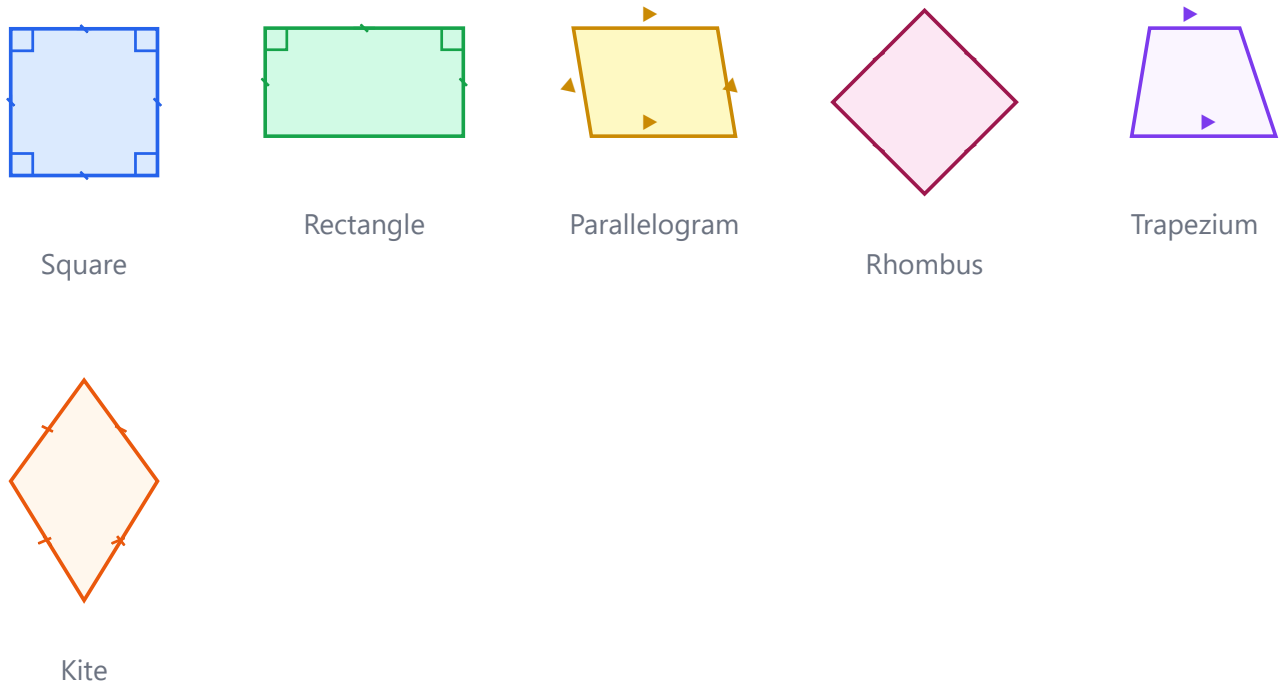
Shape	n	Interior angle sum	Each interior (regular)	Each exterior (regular)
Triangle	3	180°	60°	120°
Quadrilateral	4	360°	90°	90°
Pentagon	5	540°	108°	72°
Hexagon	6	720°	120°	60°
Octagon	8	1080°	135°	45°
Decagon	10	1440°	144°	36°

Worked Example: The interior angle of a regular polygon is 156°. How many sides does it have?

Exterior angle = $180 - 156 = 24^\circ$
$n = 360 \div 24 = \mathbf{15}$ sides

Exam Tip: If you know the exterior angle, use $n = 360 \div \text{exterior angle}$ — much quicker than using the interior angle formula.

4.1.5 Properties of Quadrilaterals



Shape	Sides	Angles	Diagonals	Lines of symmetry
Square	All 4 equal; opposite sides parallel	All 90°	Equal, bisect at 90°, bisect corner angles	4
Rectangle	Opposite sides equal and parallel	All 90°	Equal, bisect each other	2
Parallelogram	Opposite sides equal and parallel	Opposite angles equal; co-interior sum 180°	Bisect each other (not equal)	0
Rhombus	All 4 equal; opposite sides parallel	Opposite angles equal; co-interior sum 180°	Bisect at 90°, bisect corner angles (not equal)	2

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Trapezium	One pair of parallel sides	Co-interior angles between parallel sides sum to 180°	Not generally special	0 (1 if isosceles)
Kite	Two pairs of adjacent equal sides	One pair of opposite angles equal	One diagonal is perpendicular bisector of other	1

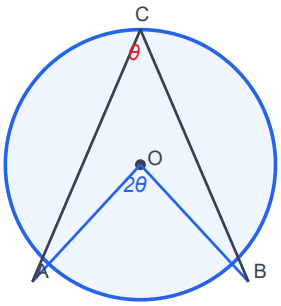
4.1.6 Circle Theorems Higher

Key vocabulary:

- **Chord** — a straight line joining two points on the circumference.
- **Arc** — part of the circumference.
- **Tangent** — a line that touches the circle at exactly one point.
- **Cyclic quadrilateral** — a quadrilateral whose four vertices all lie on a circle.
- **Subtend** — an arc or chord subtends an angle at a point if the two ends of the arc/chord are joined to that point.

Theorem 1: Angle at the Centre

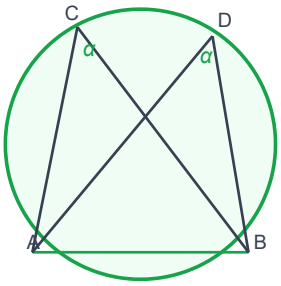
The angle subtended at the *centre* of a circle is **twice** the angle subtended at the *circumference* by the same arc.



∠AOB = 2 × ∠ACB

Theorem 2: Angles in the Same Segment

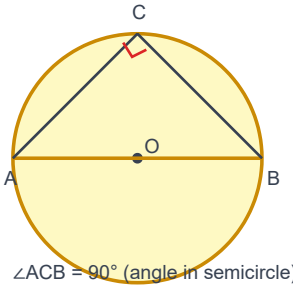
Angles subtended by the same arc (or chord) at the circumference are **equal**.



∠CAB = ∠DAB (same arc)

Theorem 3: Angle in a Semicircle

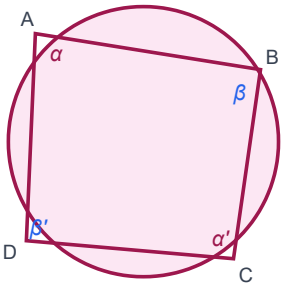
The angle subtended by a *diameter* at the circumference is always 90°.



∠ACB = 90° (angle in semicircle)

Theorem 4: Cyclic Quadrilateral

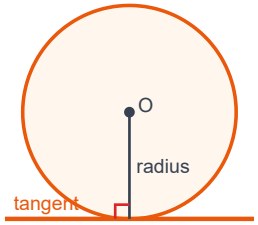
Opposite angles in a cyclic quadrilateral sum to 180° (they are *supplementary*).



α+α'=180°, β+β'=180°

Theorem 5: Tangent–Radius

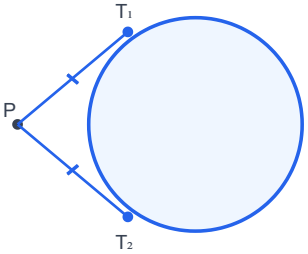
A tangent to a circle is **perpendicular** to the radius drawn to the point of tangency.



Tangent ⊥ radius at point of contact

Theorem 6: Two Tangents from an External Point

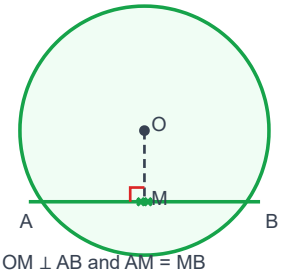
Tangent segments from an external point to a circle are **equal in length**.



$PT_1 = PT_2$

Theorem 7: Perpendicular from Centre to Chord

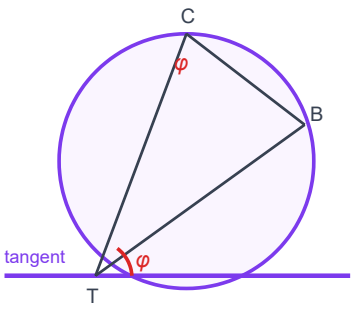
The perpendicular from the centre of a circle to a chord **bisects** the chord.



OM ⊥ AB and AM = MB

Theorem 8: Alternate Segment Theorem (Tangent–Chord Angle)

The angle between a tangent and a chord equals the inscribed angle in the *alternate segment* (the segment on the other side of the chord).



∠(tangent,chord) = inscribed angle, alt. seg

Exam Tip: When tackling circle theorem problems, mark all given angles on the diagram first. Work systematically around the circle stating each reason. Always use the full theorem name as the reason (e.g., "angle in a semicircle = 90°" not just "semicircle").

Worked Example: O is the centre of a circle. Points A, B and C lie on the circumference. Angle AOB = 130°. Find angle ACB.

Angle at centre = twice angle at circumference (same arc):
$\angle ACB = 130 \div 2 = \mathbf{65}^\circ$
Reason: angle at centre is twice angle at circumference (Theorem 1).

Worked Example: ABCD is a cyclic quadrilateral. Angle ABC = 74° and angle BCD = 110°. Find angles CDA and DAB.

Opposite angles in cyclic quadrilateral sum to 180°:
$\angle CDA = 180 - 74 = \mathbf{106}^\circ$
$\angle DAB = 180 - 110 = \mathbf{70}^\circ$

Formula Summary

Rule / Theorem	Statement
Angles on a straight line	Sum = 180°

Angles around a point	Sum = 360°
Vertically opposite	Equal
Alternate angles (parallel lines)	Equal
Corresponding angles (parallel lines)	Equal
Co-interior angles (parallel lines)	Sum = 180°
Interior angle sum of polygon	$(n-2) \times 180^\circ$
Exterior angle sum	Always 360°
Angle at centre (Th. 1)	$= 2\times$ angle at circumference
Angles in same segment (Th. 2)	Equal
Angle in semicircle (Th. 3)	$= 90^\circ$
Cyclic quadrilateral (Th. 4)	Opposite angles sum to 180°
Tangent–radius (Th. 5)	Perpendicular (90°)
Tangents from external point (Th. 6)	Equal length
Perpendicular from centre (Th. 7)	Bisects chord
Alternate segment (Th. 8)	Tangent–chord angle = inscribed angle in alt. segment

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